OMAR HAMMOUD

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EDUCATION

Swiss Federal Institute of Technology (ETH Zürich) *Zurich, Switzerland*
Master of Science in Computer Science, Data Management *Sep. 2024 - Expected early 2027*

Swiss Federal Institute of Technology Lausanne (EPFL) *Lausanne, Switzerland*
Bachelor of Science in Computer Science *Sep. 2021 - Aug. 2024*

• **Relevant Courses:** Big Data, Data Management Systems, Cloud Computing Architecture, Software Engineering, Network Security, System Security, High Performance Computing

EXPERIENCE

• **Teaching Assistant - ETH Zurich** – [Zurich, Switzerland] *Feb. 2025 - Present*
Assisted 400 students enrolled in the Databases and Software Engineering courses by preparing exercises, leading sessions, and holding office hours to help them with course projects.

• **Web Developer - HTech** – [Beirut, Lebanon] *Jul. 2024 - Oct. 2024*
Built and deployed a client's website using React and .NET, from prototype to delivery.

• **Teaching Assistant - EPFL** – [Lausanne, Switzerland] *Sep. 2023 - Jun. 2024*
Assisted 800 students enrolled in the Computer Architecture and Digital Systems courses through tutorials, lab sessions, and office hours focused on pipelining, caching, and digital logic.

PROJECTS

• **RumbleDB:** [Java] Extensive testing for the JSONiq Update Facility, measuring system performance under scaling, and implementing support for Apache Iceberg and Hudi beside Delta Lake.

• **FeedMe:** [Kotlin] Collaborated with a team of six to build a mobile app for discovering, sharing, and creating recipes, using Google Firebase for data management; ranked 2nd out of 18 teams.

• **EventLoupe:** [TypeScript, Python] Built a web app to increase student event attendance by centralizing events, showing friends' participation, and recommending events via a custom algorithm using React, FastAPI, and SQLite; placed 6th out of 27 teams at VIScon Hackathon 2025.

• **LikingNodes:** [JavaScript, Python] Developed an interactive web app where users can add and like nodes in live sessions, using `D3.js` for visualization and React, FastAPI, and WebSockets for real-time updates; deployed on Vercel, Railway, and Neon.

• **JobsOnCloud:** Evaluated static and dynamic schedulers for memcached and PARSEC workloads on Google Cloud, improving batch job completion while meeting tail-latency SLOs.

SKILLS

• **Programming Languages:**
- *Proficient in:* Java, C, Python, SQL, JavaScript
- *Familiar with:* Go, C++, Scala, Kotlin, C#, TypeScript, Assembly, Verilog, HTML, CSS

• **Technologies:** Git, Docker, Kubernetes, CI/CD, Google Cloud, Cloudflare, PySpark, ReactJS, FastAPI, Android Development, JSONiq, CUDA

• **Hobbies:** Skiing, Running, Golf, Gym

• **Languages:** Fluent in French, English and Arabic (native). Learning German